Pilot Machine Learning Model for Fuel Requirement Estimation in Agricultural Mechanization

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The Problem

Fuel estimation in Philippine agricultural mechanization relies on static coefficients.

- Generalized coefficients ignore machine variability
- No localized field, soil, or operator effects
- Nonlinear interactions invisible to deterministic formulas
- Result: weak planning, missed efficiency gains

Goal

- Develop an ML model for fuel-requirement estimation
- Compare against OLS baseline using R^2, MAE, RMSE
- Score national ABEMIS deployments for regional fuel planning
- Interpret with SHAP (global) and LIME (local)

The Data

- AMTEC test reports $\rightarrow$ **378** cleaned training records
- ABEMIS inventory $\rightarrow$ **246,741** fuel-relevant records
- Coverage: **all 17** Philippine regions
- **8** RF features: `power_kw`, year, three ABEMIS context features, three categorical-encoded (machinery type, family, subset)
- Target: fuel consumption (L/hr)

Pipeline

PDF extraction (PyMuPDF) $\rightarrow$ Cleaning + unit standardization

$\downarrow$

Feature engineering (ABEMIS context join) $\rightarrow$ Train/test split (80/20)

$\downarrow$

OLS baseline $\rightarrow$ Random Forest (8 features)

$\downarrow$

SHAP global + LIME local $\rightarrow$ ABEMIS scoring (172k records)

Methodology Highlights

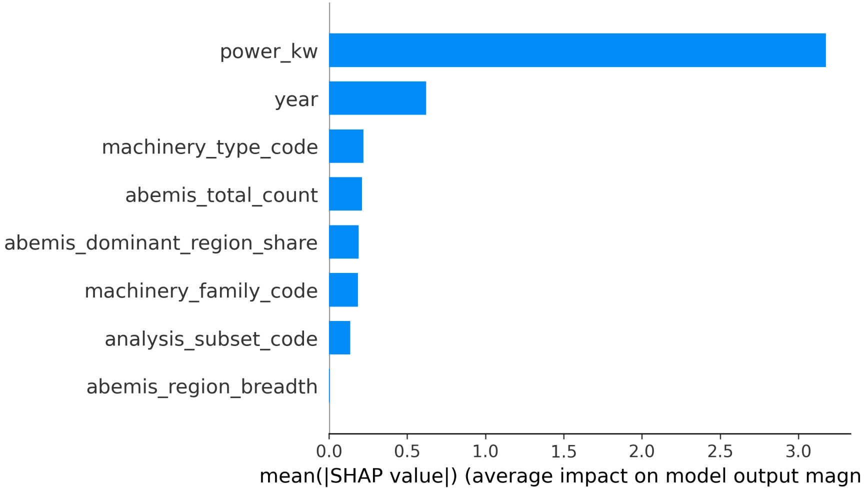
- **OLS baseline** — linear/quadratic; per-type fallback hierarchy (type $\rightarrow$ family $\rightarrow$ global)
- **Random Forest** — scikit-learn defaults, 80/20 split, 8 features, 5-fold CV
- **Interpretability** — SHAP TreeExplainer (bar + beeswarm), LIME tabular explainer (10 example records)

Results — Model Comparison ($n_{\text{test}} = 76$)

Model	**R^2**	**MAE (L/h)**	**RMSE (L/h)**
OLS Global Linear	0.761	1.683	2.640
OLS Hierarchical Fallback	0.722	1.872	2.847
Random Forest	**0.911**	**1.090**	**1.611**

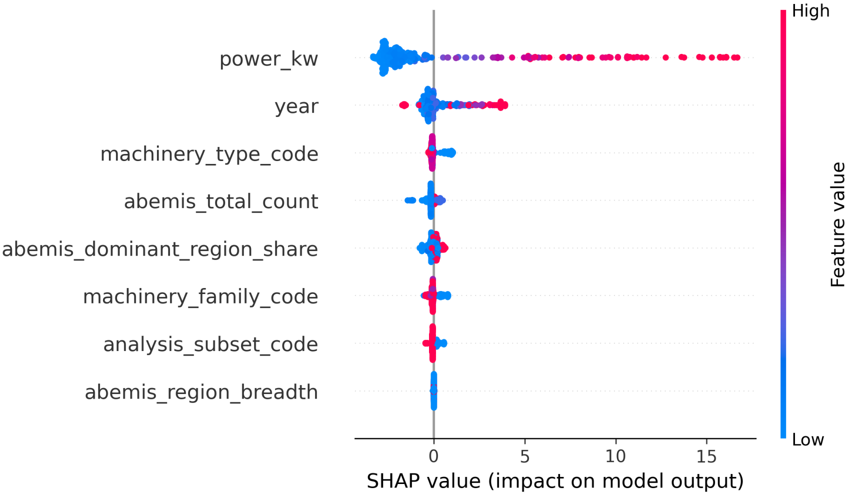
Random Forest captures non-linear power-fuel interactions OLS cannot.

SHAP — Global Feature Importance



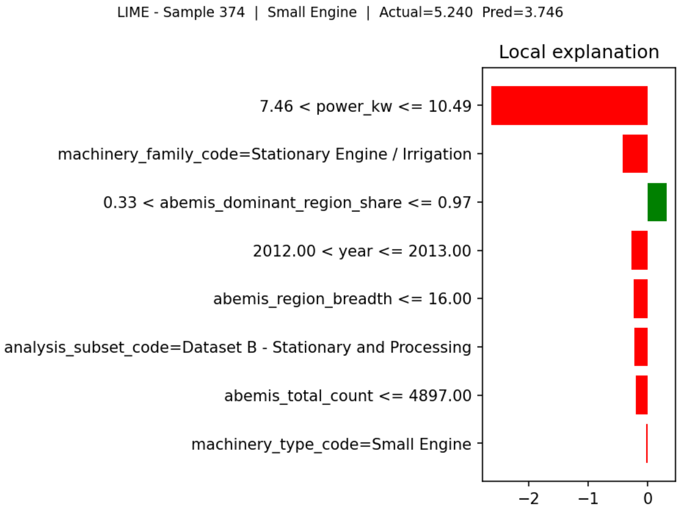
`power_kw` dominates at ∼76% importance.

SHAP — Direction & Magnitude



Higher power → higher predicted fuel; context features add nuance.

LIME — A Local Explanation



Per-record explanation builds trust for BAFE engineers.

ABEMIS Regional Scoring

- **172,265** of 246,741 records successfully scored (**70%** coverage)
- Excluded: 62,808 missing Rated Power, 11,668 unmappable type
- Outputs: per-region, per-region $\times$ type, per-record fuel estimates
- Enables data-driven mechanization planning at national scale

Limitations & Future Work

- OCR retraining on 247 scanned PDFs (Tesseract binary pending)
- RAED cropping calendar integration as the Time predictor
- Expand ABEMIS $\rightarrow$ AMTEC type mapping for $\sim$19,674 unmapped rows
- Validation against in-field telemetry from a pilot deployment

Thank you

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